

Internal Job Mobility Assistant

Career Path Intelligence and Job Matching System

Project Description:

The Internal Job Mobility Assistant harnesses Google's Gemini Generative AI to provide intelligent career mobility and job matching capabilities, offering employees a fast and personalized career growth experience. The platform includes an AI Job Matching Engine, which analyzes employee profiles against internal job openings and generates compatibility scores with detailed skill gap reports, a Career Path Generator, which delivers three personalized career path options tailored to the employee's skills and interests, and a Resume Analyzer, which extracts structured career data from raw resume text using Gemini AI. Utilizing Google's Gemini API, the platform processes structured employee profile data to deliver high-quality, context-aware AI recommendations. Built with a FastAPI backend and a React frontend styled with Tailwind CSS, the platform ensures a seamless enterprise-grade user experience. With JWT-based role authentication, SQLite database persistence via SQLAlchemy ORM, and secure API key management via .env, the Internal Job Mobility Assistant empowers employees, HR managers, and organizations to make smarter internal hiring and career development decisions with confidence.

Scenarios

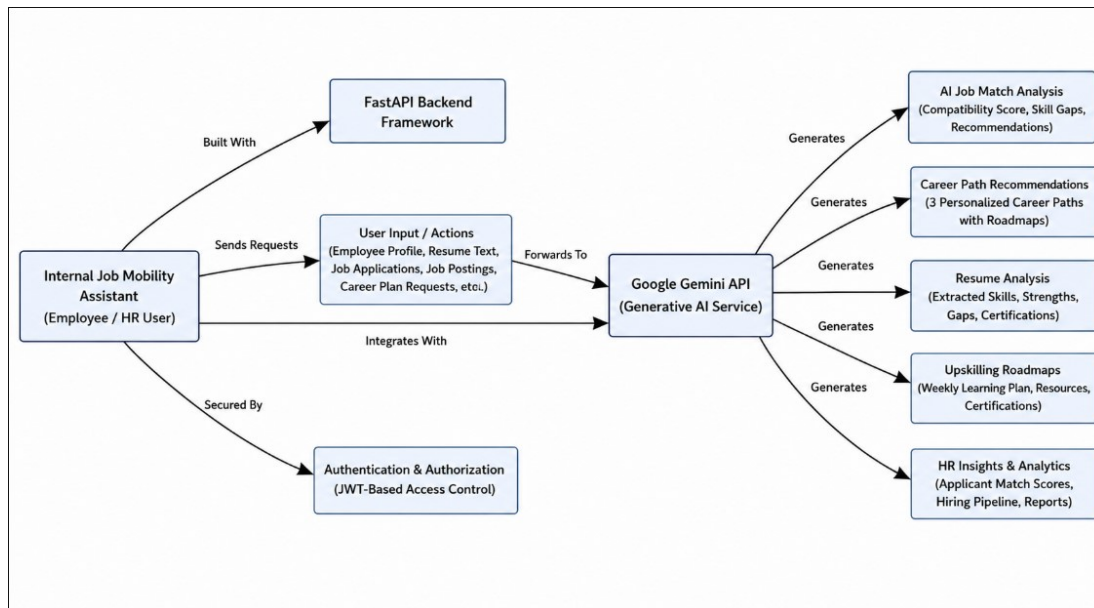
Scenario 1: An employee with Python and React skills logs in and clicks "Apply & Analyze" on a Full Stack Engineer role. The Gemini AI generates an 85% match score, explains the fit, lists missing skills like Docker, and provides a week-by-week upskilling roadmap — all in seconds.

Scenario 2: An HR manager logs in and posts a new internal vacancy for a Data Science Engineer role. They review AI match scores for all applicants, update statuses from Shortlisted to Interview to Offered, and monitor hiring analytics from the HR Business Partner Center dashboard.

Scenario 3: An employee navigates to the AI Career Plan page. Based on their current profile, Gemini generates three career paths — Upward to Senior Engineer, Lateral to Product Manager, and Upskill to AI/ML Engineer — each with a full weekly learning blueprint and recommended certifications.

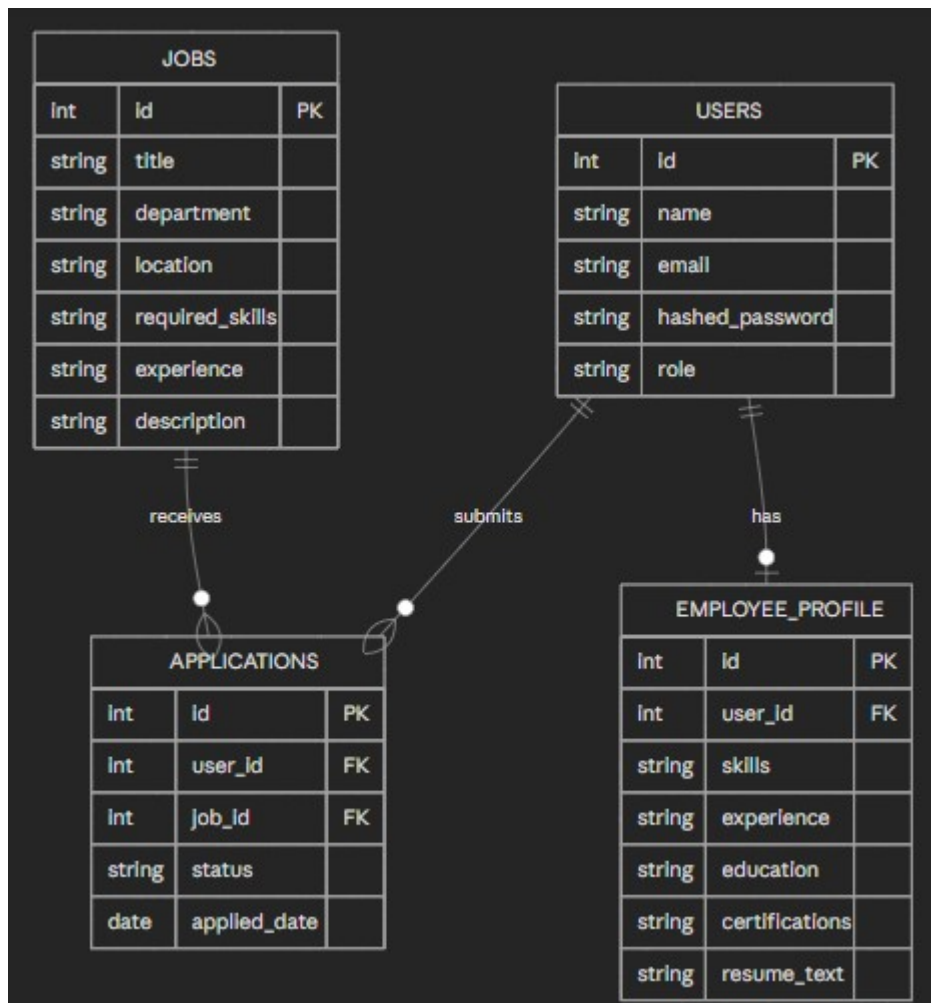
Scenario 4: An employee pastes their resume text into the Resume Analyzer. Gemini extracts their technical skills, professional strengths, development gaps, and certifications — all saved automatically to their career dossier and used immediately to improve AI job matching accuracy.

Technical Architecture:



Description:

The Internal Job Mobility Assistant utilizes a modular multi-tier architecture to deliver intelligent AI-powered career mobility and recruitment assistance. The React frontend provides a responsive and user-friendly interface where employees and HR managers can manage profiles, upload resumes, explore career opportunities, and interact with AI-powered features. The FastAPI backend handles authentication, business logic, database operations, and API request processing while securely integrating with Google's Gemini API to generate job compatibility scores, personalized career paths, resume analysis, and skill gap recommendations. Employee and organizational data are managed using SQLite with SQLAlchemy ORM, while JWT-based authentication ensures secure role-based access. The application is deployed using Vercel for the frontend and Render for the backend, providing a scalable and reliable enterprise-ready solution for internal talent mobility and career development.



Pre-requisites

Python Installation: <https://www.python.org/downloads/>
 Node.js Setup: <https://nodejs.org/en/download>
 Google Gemini API Key: <https://aistudio.google.com/>
 Visual Studio Code: <https://code.visualstudio.com/>
 FastAPI Framework : <https://fastapi.tiangolo.com/>
 React + Vite : <https://vitejs.dev/guide/>
 GitHub Version Control : <https://github.com/>

Project Workflow

Epic 1 : Gemini API Infrastructure Initialization

- Story 1 : Generate a Google Gemini API key from Google AI Studio and configure it securely.
- Story 2: Install all backend dependencies and configure the Gemini client in the AI service module.
- Story 3: Validate Gemini API connectivity by running a test generation call.

Epic 2: Core AI Logic Development

- Story 1: Build the AI job match analysis function to compare employee profiles against job requirements.
- Story 2: Implement the career path simulation function to generate three personalized career options.
- Story 3: Build the resume analyzer function to extract structured career data from resume text.
- Story 4: Add error handling, fallback responses, and JSON extraction for all Gemini API calls.

Epic 3: Backend API Development (FastAPI)

- Story 1: Design SQLAlchemy database models and configure automatic SQLite schema initialization.
- Story 2: Implement JWT authentication routes for employee and HR Partner login and registration.
- Story 3: Build REST API routes for profile management, HR job CRUD, and AI matching endpoints.
- Story 4: Configure CORS middleware and test all endpoints via FastAPI Swagger docs.

Epic 4: React Frontend Implementation

- Story 1: Build authentication pages with role selection and session restoration on page reload.
- Story 2: Build the Employee Dashboard with stat cards, dossier profile, and inline AI job analysis.
- Story 3: Build the HR Dashboard with Candidates, Vacancies, Applicants tabs and Analytics charts.
- Story 4: Build the AI Career Plan page and the Resume Analyzer page.

Epic 5: Testing and Optimization

- Story 1: Conduct end-to-end employee and HR flow testing using real profile data.
- Story 2: Validate AI output quality — match scores, skill gaps, and career roadmap structure.
- Story 3: Perform JWT security testing, API performance benchmarking, and prompt optimization.

Conclusion

➤ Conclusion

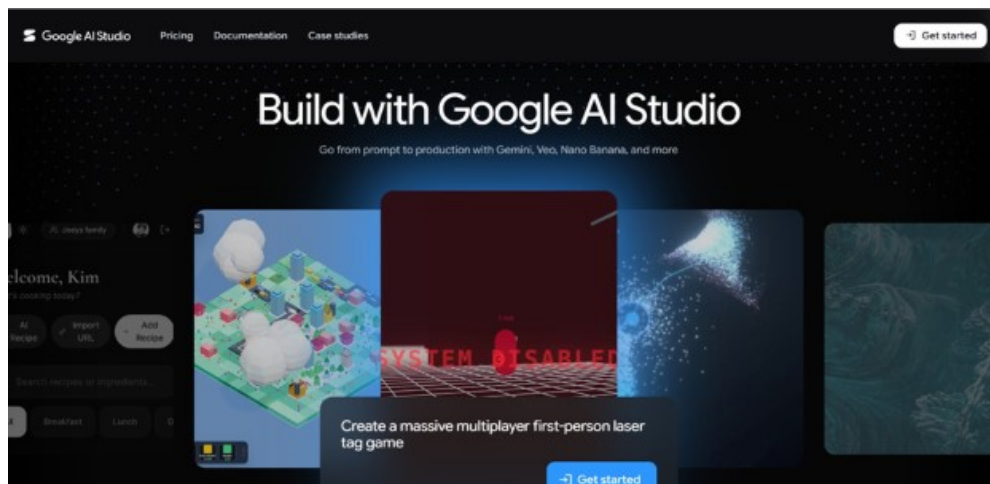
Epic 1 : Gemini API Infrastructure Initialization

In this Epic, we focus on establishing and configuring the core Gemini AI infrastructure required to power the Internal Job Mobility Assistant. This involves generating a dedicated API key from Google AI Studio, installing all backend dependencies, configuring the Gemini client securely using environment variables, and validating connectivity to ensure all AI-driven features operate seamlessly.

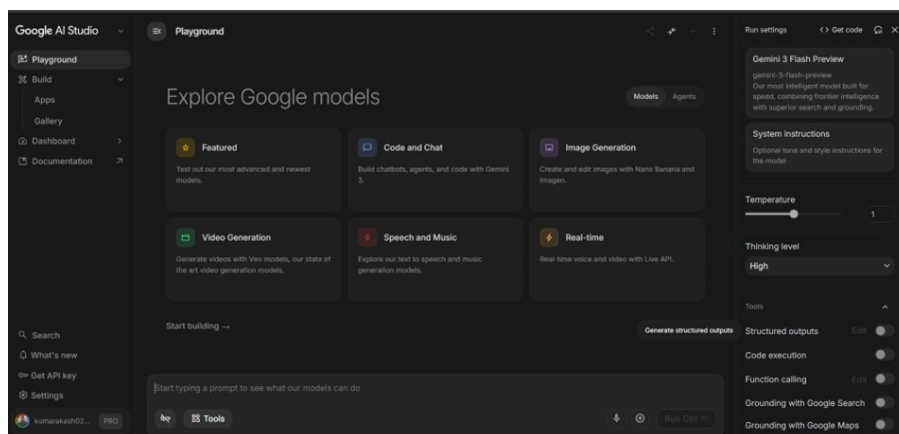
Generate a Google Gemini API Key

Before the application can communicate with the Gemini LLM, you need a valid API key. Follow the steps below to create one and connect it securely to the project.

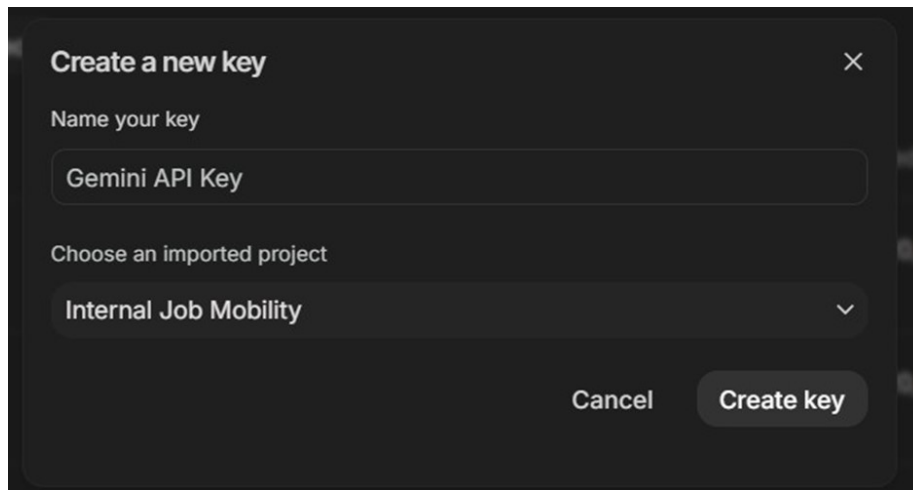
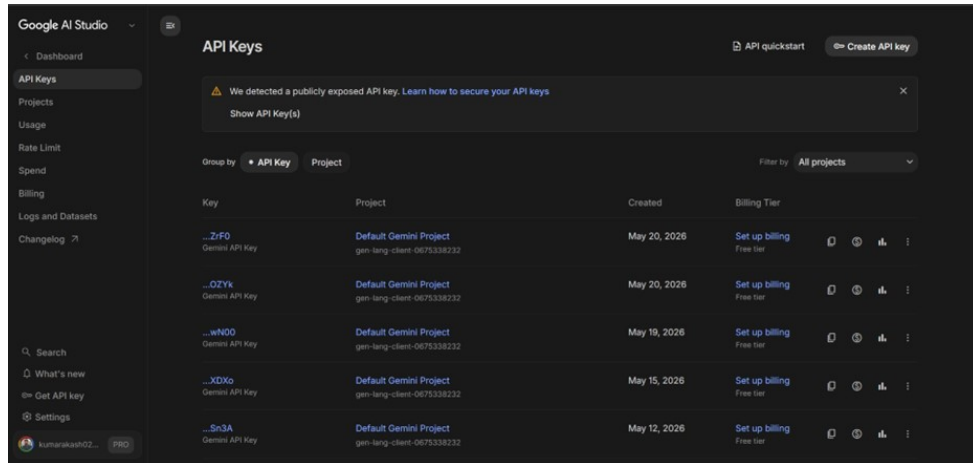
- Step 1 — Open Google AI Studio: Visit <https://aistudio.google.com/> and sign in using your Google account.



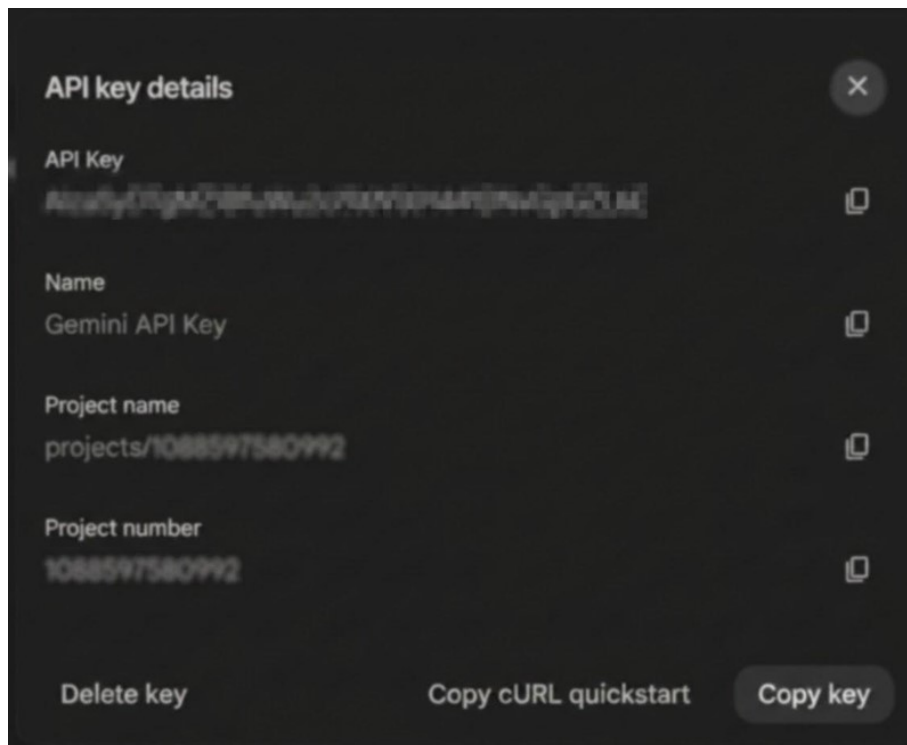
- Step 2 — Navigate to API Keys: Click on "Get API Key" in the left sidebar of the AI Studio dashboard.



- Step 3 — Create a New API Key: Click "Create API Key", select a Google Cloud project, and give your key a descriptive name such as Internal_Job_Mobility_Assistant.



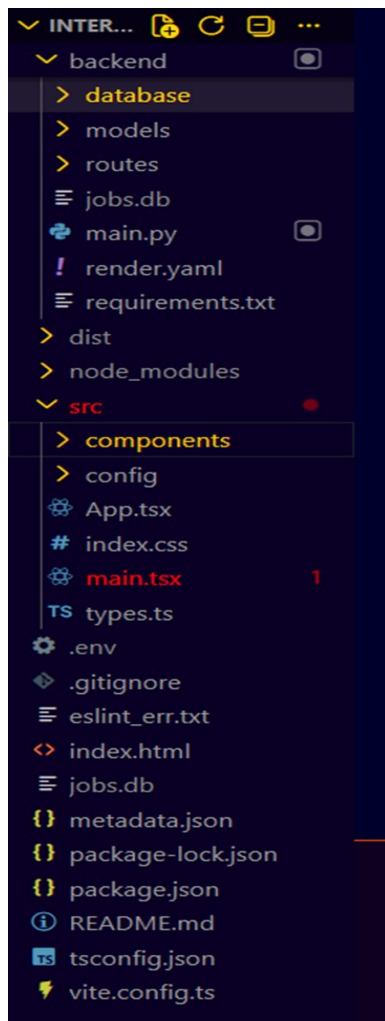
- Step 4 — Copy the API Key: Copy the key immediately after it is generated and store it safely — you will not be able to view it again after closing the dialog.



- Step 5 — Add the Key to Your Project: Create a .env file in the root of your backend folder and paste the key:
 GEMINI_API_KEY=your_copied_api_key_here
 SECRET_KEY=your_long_random_jwt_secret_here
 DATABASE_URL=sqlite:///./jobmobility.db ACCESS_TOKEN_EXPIRE_MINUTES=480
- Step 6 — Load the Key in Code: The ai_service.py file loads this key automatically at startup using os.getenv("GEMINI_API_KEY").
- Important — Never Commit Your API Key: Add .env to your .gitignore file to prevent accidentally pushing the key to a public repository.

Install Dependencies and Set Up Project Structure

- Install Backend Dependencies:
 pip install fastapi uvicorn sqlalchemy google-generativeai python-jose passlib python-dotenv pydantic python-multipart
- Install Frontend Dependencies:
 npm install
- Set Up Application Structure: Create the project folder structure as follows:



Validate Gemini LLM Connectivity

- Start the Backend Server:
`cd backend python main.py`
- Confirm the terminal shows Application startup complete without any API key warnings or errors.
- Test via the AI Status Endpoint: Navigate to <http://localhost:8000/api/ai-status> or test through the FastAPI Swagger UI at `/docs`. A successful response with "configured": true confirms Gemini is connected and all AI features are ready.

Epic 2 : Core AI Logic Development

Epic 2 focuses on building the core intelligence layer of the Internal Job Mobility Assistant. All AI logic is centralized in `backend/ai/ai_service.py` using Google's Gemini API to perform job matching, career path generation, and resume analysis.

Build the Job Match Analysis Function

Implement the `calculate_job_match_sync()` function to perform deep compatibility analysis between an employee profile and a specific job posting.

- The function receives all employee fields skills, experience, certifications, career interests — along with all job fields — title, department, requirements, required skills — and constructs a detailed prompt instructing Gemini to return a raw JSON object containing `matchPercentage`, `matchExplanation`, `skillsGap`, `recommendedCerts`, and `learningRoadmap`.

```
def analyze_job_match(employee_profile: dict, job: dict) -> dict:
    if not employee_profile.get("skills"):
        return {
            "match_score": 0,
            "match_explanation": "Employee profile is incomplete. Please add skills.",
            "skill_gaps": [],
            "upskilling_roadmap": [],
            "career_paths": [],
        }

    prompt = f"""You are a talent acquisition AI for an enterprise HR platform.

    Analyze the fit between this employee and the internal job opening.

    EMPLOYEE:
    - Current Role: {employee_profile.get('designation', 'N/A')}
    - Department: {employee_profile.get('department', 'N/A')}
    - Experience: {employee_profile.get('years_of_experience', 0)} years
    - Skills: {employee_profile.get('skills', 'N/A')}
    - Education: {employee_profile.get('education', 'N/A')}
    - Certifications: {employee_profile.get('certifications', 'N/A')}
    - Career Interests: {employee_profile.get('career_interests', 'N/A')}

    JOB:
    - Title: {job.get('title')}
    - Department: {job.get('department')}
    - Required Experience: {job.get('required_experience', 0)} years
    - Required Skills: {job.get('required_skills')}
    - Preferred Skills: {job.get('preferred_skills', 'N/A')}
    - Description: {job.get('description')}

    Respond with ONLY a JSON object, no markdown formatting, no explanation outside JSON:

    {{"match_score": 75, "match_explanation": "explanation here", "skill_gaps": ["gap1", "gap2"], "upskilling_roadmap": ["step1", "step2", "step3"], "career_paths": ["path1", "path2", "path3"]}}

    {{"match_score": 75, "match_explanation": "explanation here", "skill_gaps": ["gap1", "gap2"], "upskilling_roadmap": ["step1", "step2", "step3"], "career_paths": ["path1", "path2", "path3"]}}

    try:
        text = _call_gemini(prompt)
        return _extract_json(text)
    except Exception as e:
        print(f"[GEMINI] analyze_job_match failed: {e}")
        return {
            "match_score": 0,
            "match_explanation": "AI analysis temporarily unavailable.",
            "skill_gaps": [],
            "upskilling_roadmap": [],
            "career_paths": [],
        }
```

- If the Gemini client is unavailable, the function returns a structured fallback object with default values ensuring the platform remains functional without a live API key.

Implement the Career Path Simulation Function

Implement the `simulate_career_paths_sync()` function to generate three personalized career path options in a single Gemini API call.

- The function instructs Gemini to generate exactly three distinct career paths — each with a role title, department, transition type (Upward / Lateral / Upskill), match

percentage, explanation, skill gaps, recommended certifications, and a dynamic weekly blueprint of learning milestones.

```
def analyze_job_match(employee_profile: dict, job: dict) -> dict:
    if not employee_profile.get("skills"):
        return {
            "match_score": 0,
            "match_explanation": "Employee profile is incomplete. Please add skills.",
            "skill_gaps": [],
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            "career_paths": [],
        }

    prompt = f"""You are a talent acquisition AI for an enterprise HR platform.

    Analyze the fit between this employee and the internal job opening.

    EMPLOYEE:
    - Current Role: {employee_profile.get('designation', 'N/A')}
    - Department: {employee_profile.get('department', 'N/A')}
    - Experience: {employee_profile.get('years_of_experience', 0)} years
    - Skills: {employee_profile.get('skills', 'N/A')}
    - Education: {employee_profile.get('education', 'N/A')}
    - Certifications: {employee_profile.get('certifications', 'N/A')}
    - Career Interests: {employee_profile.get('career_interests', 'N/A')}

    ({employee_profile.get('match_score', 75), "match_explanation": "explanation here", "skill_gaps": ["gap1", "gap2"], "upskilling_roadmap": ["step1", "step2", "step3"], "ca

    try:
        text = _call_gemini(prompt)
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            "match_score": 0,
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            "skill_gaps": [],
            "upskilling_roadmap": [],
            "career_paths": [],
        }
```

Build the Resume Analyzer Function

Implement the `analyze_resume_sync()` function to extract structured career data from raw resume text.

- The function sends resume content to Gemini with an extraction prompt requesting skills, strengths, weaknesses, certifications, and an experience summary — all returned as clean JSON and saved directly to the employee's dossier in the database.

Add Error Handling and JSON Extraction

- Validate API connectivity using `check_ai_status_sync()` which sends a lightweight test prompt to Gemini and confirms a valid response before any user-facing AI feature is triggered.

```
GEMINI_MODEL = os.getenv("GEMINI_MODEL", "gemini-2.5-flash")
```

```
def get_genai_client():
    api_key = os.getenv("GEMINI_API_KEY")
    if not api_key or api_key == "MY_GEMINI_API_KEY":
        return None
    try:
        return genai.Client(api_key=api_key)
    except Exception as e:
        print(f"Error initializing GenAI Client: {e}")
        return None
```

- All service functions use try-except around JSON parsing to handle cases where Gemini wraps responses in markdown code fences. A structured fallback object is always returned on failure, preventing frontend crashes.

Epic 3 : Backend API Development (FastAPI)

Epic 3 focuses on building the complete REST API backend. The goal is to implement secure, modular endpoints for authentication, employee profile management, HR job posting, and AI integration.

Database Models and Schema Initialization

Define SQLAlchemy ORM models for all database tables — DBUser, DBEmployeeProfile, DBInternalJob, DBRecommendation, DBResume, and DBApplication — and configure automatic SQLite schema creation on application startup using Base.metadata.create_all(bind=engine).

```
import datetime
from sqlalchemy import Column, String, Integer, Text, Table, ForeignKey, Float, DateTime
from sqlalchemy.orm import declarative_base, relationship

Base = declarative_base()

class DBUser(Base):
    __tablename__ = "users"

    id = Column(String, primary_key=True, index=True)
    name = Column(String, nullable=False)
    email = Column(String, unique=True, index=True, nullable=False)
    hashed_password = Column(String, nullable=False)
    role = Column(String, default="employee") # "employee" or "hr"

    profile = relationship("DBEmployeeProfile", uselist=False, back_populates="user")

class DBEmployeeProfile(Base):
    __tablename__ = "employee_profiles"

    id = Column(String, primary_key=True, index=True)
    user_id = Column(String, ForeignKey("users.id"), unique=True, nullable=False)
    skills = Column(Text, default="") # Comma separated skilled items
    experience = Column(Text, default="")
    certifications = Column(Text, default="") # Comma separated certifications
    career_interests = Column(Text, default="")
    resume_text = Column(Text, nullable=True)
    resume_name = Column(String, nullable=True)
```

```
class DBEmployeeProfile(Base):
    user = relationship("DBUser", back_populates="profile")

class DBInternalJob(Base):
    __tablename__ = "internal_jobs"

    id = Column(String, primary_key=True, index=True)
    title = Column(String, nullable=False)
    department = Column(String, nullable=False)
    location = Column(String, nullable=False)
    salary_range = Column(String, nullable=True)
    description = Column(Text, nullable=False)
    requirements = Column(Text, default="") # Newline separated list of credentials
    skills_needed = Column(Text, default="") # Comma separated list of skills
    posted_date = Column(String, nullable=False)
    experience_level = Column(String, nullable=True) # e.g. Junior, Mid, Senior, Lead
    employment_type = Column(String, nullable=True) # e.g. Full-time, Permanent, Contract

class DBRecommendation(Base):
    __tablename__ = "recommendations"

    id = Column(String, primary_key=True, index=True)
    user_id = Column(String, nullable=False)
    job_id = Column(String, nullable=False)
    match_percentage = Column(Integer, default=0)
    match_explanation = Column(Text, nullable=False)
```

Implement JWT Authentication Routes

Implement `/api/auth/register` and `/api/auth/login` endpoints with `bcrypt` password hashing and JWT token generation. The `get_current_user_jwt()` dependency is injected into all protected routes to automatically validate the Bearer token on every request.

```
import datetime
from typing import Optional
import jwt
from passlib.context import CryptContext
from fastapi import Depends, HTTPException, status
from fastapi.security import HTTPBearer, HTTPAuthorizationCredentials

SECRET_KEY = "aura-secret-mobility-token-key-2026"
ALGORITHM = "HS256"

pwd_context = CryptContext(schemes=["bcrypt"], deprecated="auto")
security_scheme = HTTPBearer()

def hash_password(password: str) -> str:
    return pwd_context.hash(password)

def verify_password(plain_password: str, hashed_password: str) -> bool:
    return pwd_context.verify(plain_password, hashed_password)

def create_access_token(data: dict, expires_delta: Optional[datetime.timedelta] = None) -> str:
    to_encode = data.copy()
    if expires_delta:
        expire = datetime.datetime.utcnow() + expires_delta
    else:
        expire = datetime.datetime.utcnow() + datetime.timedelta(days=7)
    to_encode.update({"exp": expire})
    encoded_jwt = jwt.encode(to_encode, SECRET_KEY, algorithm=ALGORITHM)
    return encoded_jwt

def get_current_user_jwt(credentials: HTTPAuthorizationCredentials = Depends(security_scheme)) -> dict:
    token = credentials.credentials
    try:
        payload = jwt.decode(token, SECRET_KEY, algorithms=[ALGORITHM])
        return payload
    except jwt.PyJWTError:
        raise HTTPException(
            status_code=status.HTTP_401_UNAUTHORIZED,
            detail="Could not validate credentials",
            headers={"WWW-Authenticate": "Bearer"},
        )
```


Build Employee, Job, and AI Routes

Define all application endpoints:

- GET/PUT /api/profile — fetch and update employee career dossier
- POST /api/resume/analyze — trigger Gemini resume extraction
- GET/POST/DELETE /api/jobs — list, create, and remove job postings
- POST /api/jobs/match — trigger AI job match analysis
- POST /api/recommendations/generate — batch generate career paths
- GET /api/ai-status — check Gemini API connectivity

Configure CORS and Test APIs

Configure CORS middleware to allow React frontend communication and test all endpoints via the FastAPI Swagger UI at <http://localhost:8000/docs>.

Epic 4 : React Frontend Implementation

Epic 4 focuses on building the complete React frontend. The objective is to deliver a modern, responsive, enterprise-grade UI for both Employee and HR roles — styled with Tailwind CSS and connected to the FastAPI backend via the Fetch API with Bearer token authorization.

Build Authentication Pages and Session Management

Implement LoginRegister.tsx with role selection (Employee / HR Partner), form validation, and JWT token storage in localStorage. In App.tsx, implement restoreSession() to read the saved token on page load, validate it via /api/auth/me, and restore the user session without requiring re-login.

Build Employee Dashboard and Career Dossier

Build the Employee Dashboard with four stat cards, an AI-Recommended Roles section powered by Gemini, a Career Dossier profile with skills tags and certifications, and inline AI analysis display within job cards showing match score, skill gaps, and learning roadmap after clicking "Apply & Analyze".

Build HR Dashboard with Analytics

Build HRDashboard.tsx with three tabs — Candidates, Vacancies, and Applicants — and a Post New Vacancy modal. Build AnalyticsPage.tsx featuring four Recharts visualizations: a bar chart for in-house skill inventories, a donut chart for positions by division, a bar chart for match score distributions, and a line chart for mobility engagement.

Build AI Career Plan and Resume Analyzer Pages

Build `AIRecommendations.tsx` displaying three Gemini-generated career path cards each with a full Dynamic Weekly Blueprint roadmap and a Regenerate button. Build `ResumeUpload.tsx` with a resume text input area, Gemini AI Insights Report panel, and an "Apply Details to Dossier" button that saves extracted data to the employee profile.

Epic 5 : Testing and Optimization

In Epic 5, the focus is on validating the reliability, accuracy, and AI output quality of the Internal Job Mobility Assistant across all features, using real employee profiles and job postings.

End-to-End Flow Testing

Employee Flow Testing: Register as an employee, complete the career dossier, browse jobs, click "Apply & Analyze", and verify the AI generates a meaningful match score, skill gap list, and roadmap that accurately reflects the profile-to-job comparison.

HR Flow Testing: Register as an HR Partner, post internal vacancies, review applicant AI match scores, update application statuses, and verify the Analytics dashboard charts update correctly.

AI Output Quality Validation

Match Score Grounding Test: Verify the generated outputs reflect realistic compatibility scores, provide meaningful explanation text aligned with the profile and job data, correctly identify missing skills, and generate relevant upskilling steps specific to the skill gaps.

Career Roadmap Quality Test: Validate that the three career paths contain realistic transition types, achievable short-term goals, role-relevant skill gaps, and specific weekly learning milestones in the roadmap content.

Performance and Security Optimization

JWT Authentication Testing: Verify token persistence across browser refresh, confirm logout clears the session, and validate that role-based access control prevents employees from accessing HR-only endpoints.

API Performance Testing: Confirm response times — profile update under 200ms, job listing under 300ms, AI match analysis 3–8 seconds, and career path generation 4–10 seconds.

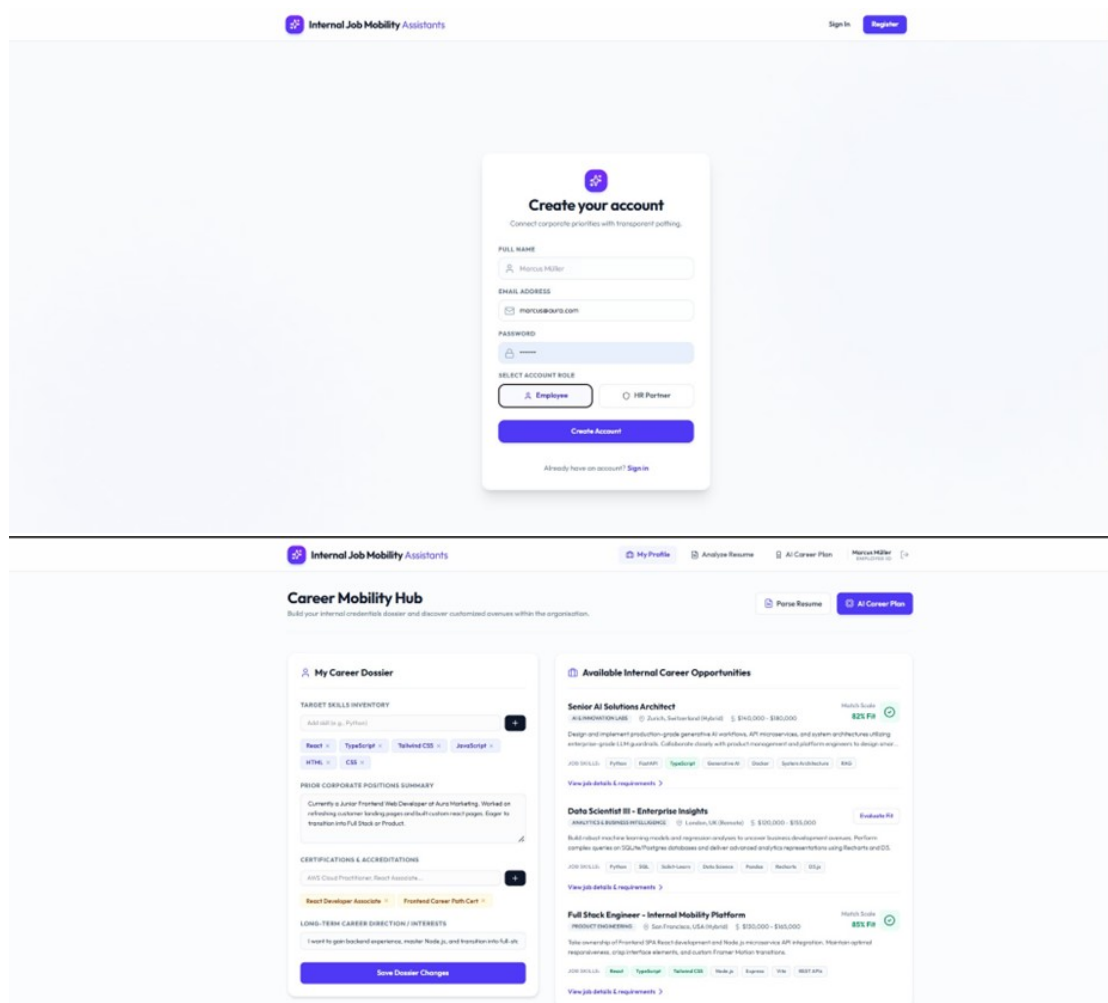
Prompt Optimization: Refine JSON instruction clarity, improve profile data formatting for better AI comprehension, and adjust retry delay intervals for rate limit handling on free tier API keys.

Important Notes:

- The Gemini API free tier has rate limits. If 429 errors occur during testing, wait 60 seconds before retrying.
- Ensure the .env file is added to .gitignore before the first GitHub push.
- All AI response times depend on Gemini API server load and may vary between 2–15 seconds on free tier keys.

Exploring the Website's Web Pages:

Employee Career Mobility Hub:



The screenshot displays the 'Internal Job Mobility Assistants' web application. At the top, there is a navigation bar with 'Sign In' and 'Register' buttons. The main content area is divided into two sections. The upper section, titled 'Create your account', contains a form with fields for 'Full Name', 'Email Address', and 'Password', along with a 'Select Account Role' dropdown (options: Employee, HR Partner) and a 'Create Account' button. The lower section, titled 'Career Mobility Hub', features a 'My Career Dossier' on the left and 'Available Internal Career Opportunities' on the right. The 'My Career Dossier' includes sections for 'Target Skills Inventory', 'Prior Corporate Positions Summary', 'Certifications & Accreditations', and 'Long-Term Career Direction / Interests'. The 'Available Internal Career Opportunities' section lists three job openings: 'Senior AI Solutions Architect', 'Data Scientist III - Enterprise Insights', and 'Full Stack Engineer - Internal Mobility Platform', each with a brief description, location, salary range, and a 'Match Score'.

Description: The Employee Career Mobility Hub is the central dashboard for logged-in employees. It displays four metric stat cards at the top — Open Positions, Active Applications, Top Match %, and Profile Completion % — followed by a Career Dossier panel on the left for managing skills, experience, and certifications, and an Available Internal Career Opportunities panel on the right showing all job openings with AI-calculated match scores and an "Apply & Analyze with AI" button on each card.

AI Analysis Result:

TARGET REVOLUTION BLUEPRINT

Junior Full Stack Developer

Personalized week-by-week upskilling map generated to bridge specific organizational standards.

UNRESOLVED SKILL GAPS

- Node.js ecosystem (Express.js, NestJS)
- Database management (SQL: PostgreSQL/MySQL; NoSQL: MongoDB)
- RESTful API design and implementation
- Backend testing (unit, integration)
- Authentication and Authorization mechanisms
- Deployment strategies (Docker, cloud platforms like AWS/Azure)

KEY CREDENTIAL RECOMMENDATIONS

- Node.js & Express.js Course Completion (e.g., Udemy, Coursera)
- MongoDB Developer Certification (Associate) or SQL Fundamentals Course

DYNAMIC WEEKLY BLUEPRINT MILESTONES

```

'''
markdown
**Week 1-4: Node.js & Express.js Fundamentals**
* Deep dive into Node.js runtime, modules, event loop, and async programming.
* Learn Express.js for building RESTful APIs, routing, and middleware.
* Build simple CRUD APIs without a database, focusing on request/response cycles.
* Familiarize with Postman or Insomnia for API testing.
**Week 5-8: Database Integration & ORMs**
* Introduction to SQL (PostgreSQL/MySQL) and NoSQL (MongoDB) concepts.
* Learn an ORM/ODM (e.g., Sequelize for SQL, Mongoose for MongoDB) for data interaction.
* Integrate a chosen database into existing Node.js APIs.
* Practice data modeling, migrations, and complex querying.
**Week 9-12: Authentication, Authorization & Deployment Basics**
* Implement user authentication (e.g., JWT, Passport.js) and authorization strategies.
* Containerization with Docker for Node.js applications.
* Basic deployment to a cloud platform (e.g., Heroku, Vercel for backend, AWS EC2/Lambda).
* Start a small full-stack internal project combining new backend skills with existing React frontend expertise.
**Week 13-16: Advanced Concepts & Project Contribution**
* Implement robust error handling, logging, and performance monitoring for backend services.
* Write unit and integration tests for backend APIs using Jest or Supertest.
* Contribute to an existing internal backend service or lead a new full-stack feature under mentorship.
* Regular pair programming sessions with senior full-stack developers to learn best practices and architecture.
'''

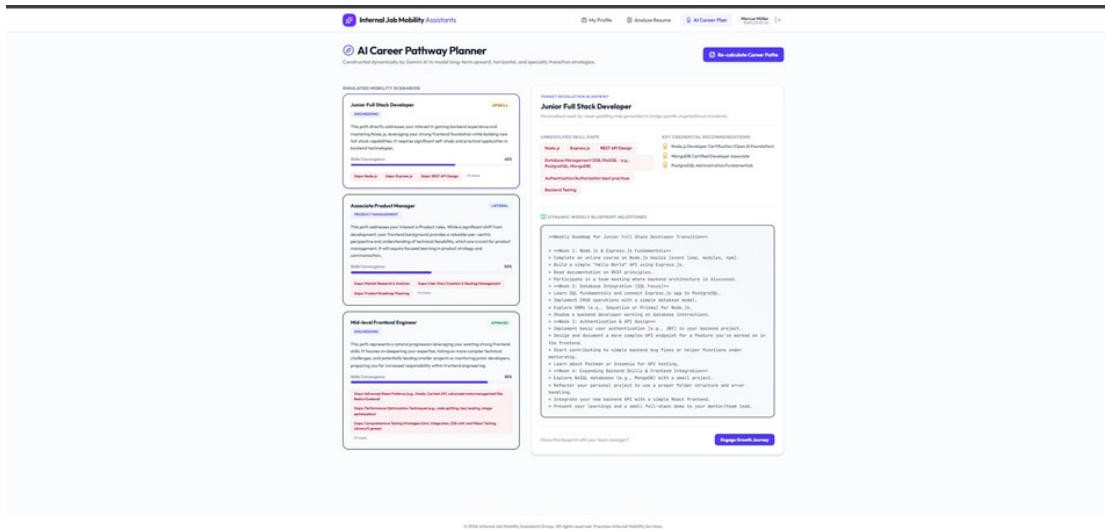
```

Share this blueprint with your team manager?

Engage Growth Journey

Description: When an employee clicks "Apply & Analyze", the full Gemini AI analysis appears inline within the job card. It displays a colored match score badge, a detailed explanation of the compatibility assessment, a Skill Gaps list identifying missing skills, Recommended Certifications to bridge the gaps, and a structured Learning Roadmap providing a week-by-week upskilling plan specific to the target role.

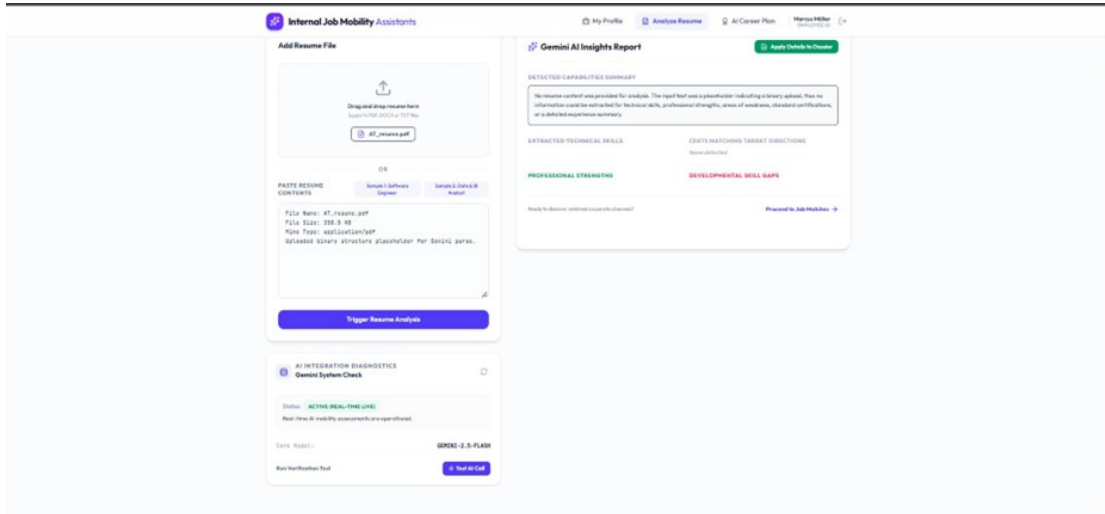
AI Career Plan Page:



The screenshot shows the 'AI Career Pathway Planner' interface. It features a navigation bar with 'Internal Job Mobility Assistants', 'My Profile', 'Analysis Resume', 'AI Career Plan', and 'Marissa Miller'. The main content area is titled 'AI Career Pathway Planner' and includes a sub-header 'Continuously generated by Gemini AI technology from up-to-date, verified, and quality information on the job market'. A 'Regenerate Career Paths' button is visible. The interface displays three career path options for a 'Junior Full Stack Developer' role, each with a 'Match' percentage and a 'Transition Type' (Upward / Lateral / Upskill). The paths are: 'Junior Full Stack Developer' (Match: 85%, Transition Type: Upward), 'Associate Product Manager' (Match: 80%, Transition Type: Lateral), and 'Mid-Level Product Engineer' (Match: 85%, Transition Type: Upskill). Each path includes a 'Dynamic Weekly Blueprint Milestones' section with a list of tasks and milestones. A 'Regenerate Career Paths' button is located at the bottom right of the main content area.

Description: The AI Career Pathway Planner presents three Gemini-generated career path options tailored to the employee's current profile. Each card displays the target role, transition type (Upward / Lateral / Upskill), match percentage, explanation, skill gaps, recommended certifications, and a full Dynamic Weekly Blueprint Milestones roadmap. A "Regenerate Career Paths" button triggers a fresh Gemini API call whenever the employee wants updated recommendations.

Resume Analyzer Page:



The screenshot shows the 'Resume Analyzer' interface. It features a navigation bar with 'Internal Job Mobility Assistants', 'My Profile', 'Analysis Resume', 'AI Career Plan', and 'Marissa Miller'. The main content area is titled 'Resume Analyzer' and includes a sub-header 'Gemini AI Insights Report'. A 'Trigger Resume Analysis' button is visible. The interface displays a 'Detected Capabilities Summary' section with a list of skills and strengths. Below this, there are sections for 'Extracted Technical Skills', 'Professional Strengths', 'Developmental Skill Gaps', and 'Recommended Certifications'. An 'AI Integration Diagnostics' section at the bottom confirms the Gemini system is active. An 'Apply Details to Dossier' button is located at the bottom right of the main content area.

Description: The Resume Analyzer page allows employees to paste resume text and receive a Gemini-powered extraction report. The AI Insights Report panel displays extracted technical skills, professional strengths, developmental skill gaps, and recommended certifications. An AI Integration Diagnostics section confirms Gemini is connected and active. An "Apply Details to Dossier" button saves all extracted data to the employee's career profile for immediate use in job matching.

HR Business Partner Center:

Internal Job Mobility Assistants

Talent Hub Global Analytics Sabine Kaufmann HR BUSINESS PARTNER

Senior HR Business Partner Center

Supervise internal candidates matching indexes, aggregate division talent inventories, and authorize new positions.

View Analytics Dashboard Post New Vacancy

CANDIDATES (2) VACANCIES (2) APPLICANTS (2) Filter candidates or roles...

Marcus Müller
marcusmuller.com

PROFESSIONAL SUMMARY & TALENTS
Currently a Senior Frontend Web Developer at SmartBridge. Worked on refactoring customer landing pages and built custom react pages. eager for transition into full stack or product.

Recent Division Vacancies: Frontend/Backend Web Dev

SKILL INDEX TRENDS: React, TypeScript, JavaScript, HTML, CSS

Elena Petrova
elena@smart.com

PROFESSIONAL SUMMARY & TALENTS
10+ years of experience handling metrics dashboards. Highly analytical, comfortable building dashboards, and writing robust SQL queries.

Recent Division Vacancies: Data Analyst, Product Manager, Data Scientist

SKILL INDEX TRENDS: Python, SQL, Product, Analytics, Data Science

Strategic Mobility Planning

Target horizontal internal transitions to retain intellectual property, close vacant critical skills quickly, and reduce recruitment spend.

- 92% of transitioned roles report higher output metrics in their first year.
- Recalculate models globally on the Analytics dashboard page anytime.

Internal Job Mobility Assistants

Talent Hub Global Analytics Sabine Kaufmann HR BUSINESS PARTNER

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Supervise internal candidates matching indexes, aggregate division talent inventories, and authorize new positions.

View Analytics Dashboard Post New Vacancy

CANDIDATES (2) VACANCIES (4) APPLICANTS (1) Filter applicant name or title...

Applicant ID: app_v1chf8f 2024-05-24

Marcus Müller
marcusmuller.com

Applied For: Senior AI Solutions Architect - AI & Innovation Labs

STATUS: OFFERED

Shortlist Interview Offer Reject

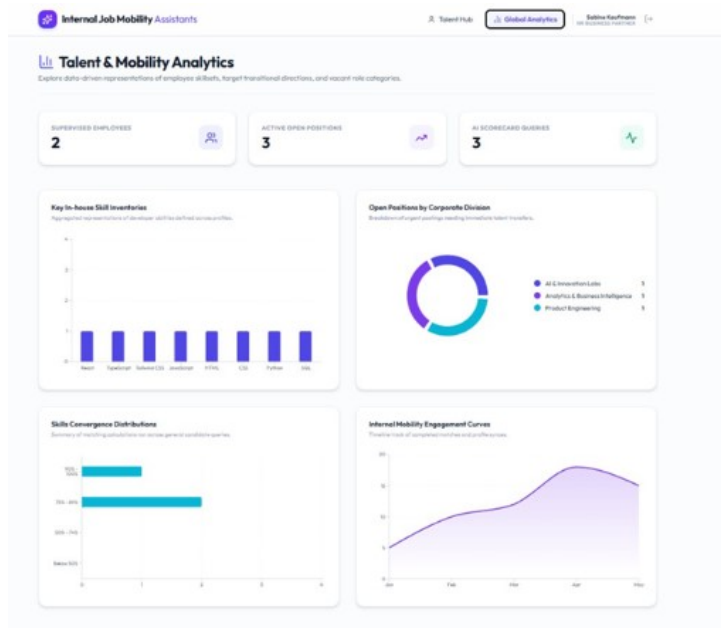
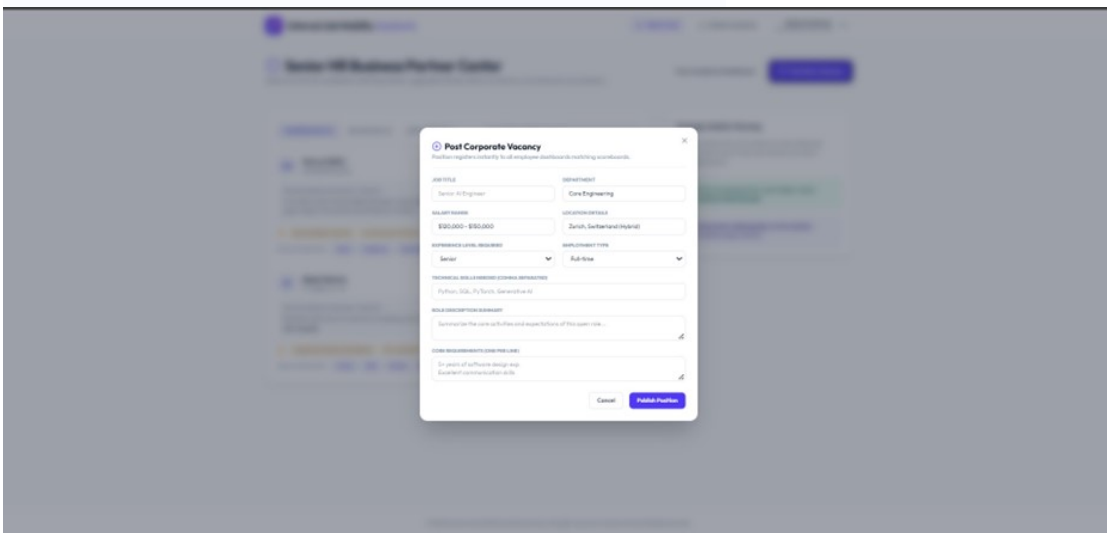
Strategic Mobility Planning

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Description: The Senior HR Business Partner Center provides the HR Partner with a complete talent management interface. The Candidates tab shows all registered employees with profile summaries. The Vacancies tab displays all posted jobs with edit and delete controls plus a Post New Vacancy modal. The Applicants tab shows all submitted applications with AI match scores and status management. A Strategic Mobility Planning panel on the right provides key organizational insights.

Global Analytics Page:

Post Corporate Vacancy
Position requires review by full employee dashboard matching requirements.

Job Title: Senior AI Engineer
Department: Core Engineering

Salary Range: \$20,000 - \$50,000
Location: Zurich, Switzerland (remote)

Experience Level: Senior
Employment Type: Full-time

Required Skills/Qualifications:
Python, SQL, PyTorch, Generative AI

Additional Information:
Summarize the core skills and qualifications of the open role.

Code Requirements (optional):
Exposure to software design and development in AI/ML.

Buttons: Cancel, Publish Position

Description: The Talent & Mobility Analytics page provides HR teams with a comprehensive data visualization dashboard. Three metric cards at the top show supervised employees count, active open positions, and AI scorecard queries. Below are four interactive Recharts charts — a bar chart of in-house skill inventories, a donut chart of open positions by corporate division, a bar chart of AI match score distributions across all applications, and a line chart tracking internal mobility engagement over six months.

Conclusion

This is the final section of a project report that summarizes the key work completed, highlights the main outcomes and findings, and provides closure to the document by reflecting on the project's overall success .

Conclusion

The Internal Job Mobility Assistant is a Generative AI-powered platform designed to enhance internal career development and talent management within organizations. Built using FastAPI and React, the system helps employees discover suitable internal opportunities by analyzing their profiles, skills, experience, and career aspirations. By leveraging Google's Gemini API, the platform delivers intelligent job matching, identifies skill gaps, and provides personalized recommendations that support informed career growth decisions.

The application integrates advanced features such as AI-driven compatibility scoring, customized upskilling roadmaps, career path recommendations, resume analysis, JWT-based authentication, and role-based dashboards. Interactive analytics and visualizations implemented with Recharts enable both employees and HR professionals to gain actionable insights into workforce capabilities, talent readiness, and career progression opportunities. This comprehensive feature set creates a seamless and data-driven internal mobility experience.

Overall, the project demonstrates how Generative AI can transform traditional HR processes by connecting employee potential with organizational opportunities more effectively. Its modular architecture, secure implementation, and production-ready design make it suitable for portfolio showcases, academic projects, and future enterprise deployment. With further enhancements and scalability improvements, the Internal Job Mobility Assistant has the potential to become a powerful workforce development and talent optimization solution.